# Literature Review and Project Rationale

Real-Time Supply Chain Monitoring Platform

## Purpose of the Literature Review

The purpose of this literature review is to establish the conceptual and technical foundation for the proposed Real-Time Supply Chain Monitoring Platform. It examines relevant research on supply chain visibility, data-driven decision-making, Big Data Analytics, and the processing of operational data. The review is intended to inform the project's requirements, justify its general technical direction, and identify the practical considerations that should guide its development.

Rather than attempting to develop a new supply chain theory or claim a scientific breakthrough, the review focuses on understanding established approaches and translating relevant findings into a practical data engineering project. Particular attention is given to the relationship between reliable data processing and the ability to monitor shipments, analyze delivery delays, and observe inventory changes.

## Background and Existing Research

### Supply Chain Visibility and Operational Monitoring

Supply chain visibility refers to the ability to obtain and interpret information about the movement of goods and the state of operations across a supply chain. Limited visibility can make it difficult to identify delayed shipments, understand changes in inventory, and recognize operational problems in a timely manner.

Koot, Mes, and Iacob (2021) reviewed research on supply chain decision-making supported by the Internet of Things (IoT) and Big Data Analytics. Their review describes how data collected from physical operations can be combined with analytical systems to improve monitoring and support operational decisions. It also highlights the importance of connecting data acquisition with processing and analysis rather than treating data collection as an isolated activity [1].

Taj et al. (2023) examined IoT-based supply chain management, including tracking, traceability, and the technologies used to support supply chain operations. Their review provides relevant context for understanding how monitoring systems can improve the visibility of goods as they move through different stages of a supply chain [2].

These studies establish the relevance of timely and accessible operational information. For the proposed project, this translates into requirements for structured shipment records, meaningful status information, and indicators that help users interpret delays and inventory changes.

### Big Data Analytics in Supply Chain Management

Supply chain operations can generate substantial amounts of data from transportation, inventory management, warehousing, and other business activities. Analyzing these data can help organizations identify patterns, evaluate performance, and make more informed decisions.

Lee and Mangalaraj (2022) reviewed Big Data Analytics research in supply chain management from both organizational and technical perspectives. Their study discusses analytical approaches,

technical methods, and the potential relationship between data analytics and supply chain performance [3].

However, the usefulness of analytics depends on the quality and accessibility of the underlying data. Inconsistent records, invalid values, duplicated events, and unreliable processing can undermine analytical results. Consequently, a monitoring platform should address data preparation, validation, storage, and processing alongside visualization.

This consideration directly informs the proposed project's emphasis on data quality checks, repeatable pipeline execution, database consistency, and the validation of results across processing stages.

## Relevance to the Proposed Project

The reviewed literature provides a basis for several aspects of the proposed platform.

- **Shipment visibility:** Organizing shipment information so that status changes and potential delivery delays can be examined.
- **Inventory monitoring:** Presenting inventory-related information in a structured form to support operational analysis.
- **Data quality and consistency:** Applying validation and cleaning procedures before data are used for reporting and analysis.
- **Scalable data processing:** Introducing distributed processing techniques as a later development stage to demonstrate how larger datasets can be handled.
- **Operational reporting:** Presenting selected indicators through a dashboard that makes processed information easier to interpret.

The literature supports the general importance of these capabilities. The specific implementation choices described below are proposed engineering decisions for this project, not claims that the reviewed studies prescribe one mandatory technology stack.

## Proposed Technical Direction

The platform is planned as a staged data engineering solution. Each stage builds on the preceding one, allowing the core data workflow to be established before optional or more complex components are introduced.

### Initial Batch Processing Stage

The initial stage is planned to use Python and Pandas for data ingestion, cleaning, transformation, and validation. SQL Server will provide relational storage for structured operational data. This stage will establish the baseline processing workflow and the rules required to maintain consistent records.

The pipeline is also expected to address duplicate handling, record updates, error logging, and data quality checks. These capabilities are important because monitoring results are only useful when the underlying records can be processed consistently and their quality can be assessed.

### Big Data Processing Stage

A subsequent stage is planned to introduce HDFS, Apache Spark through PySpark, and Parquet storage. These technologies will be used to explore distributed storage and processing, as well as a columnar file format suitable for analytical workloads.

The project will compare relevant outputs between the relational and distributed processing stages to assess consistency. The comparison will depend on the dataset and validation rules selected during implementation.

Partitioning and bucketing may also be considered where they are appropriate to the data model and query patterns. Their use should be justified by the implementation rather than included solely to increase the number of technologies in the project.

### Monitoring and Event Simulation

A Streamlit dashboard is planned to present selected shipment, delay, and inventory indicators. To demonstrate changes over time without requiring physical tracking devices, the project may use simulated events or replayed records.

Kafka and Airflow are considered optional extensions, subject to the available time and successful completion of the core pipeline. Their inclusion is not a prerequisite for demonstrating the initial batch workflow.

The term *real-time* in the project title therefore describes the intended monitoring objective. Any demonstration based on simulated events will be identified as simulated, and its actual processing behavior will be evaluated during implementation.

## Practical Considerations and Project Scope

The literature identifies the value of integrating operational data with analytical capabilities, but a practical implementation must also account for its own data, infrastructure, and time constraints.

The project will therefore distinguish between established background concepts and features that remain to be implemented and tested. The final dataset, available fields, and data quality characteristics will determine which operational indicators can be calculated reliably.

The initial scope does not include direct integration with production IoT devices or live logistics infrastructure. Instead, the project focuses on demonstrating the main data engineering stages using an available dataset and, where appropriate, simulated events.

The four-week project timeline also requires prioritization. The essential batch pipeline, relational storage, data validation, and core documentation take priority over optional orchestration or streaming components. This approach keeps the project feasible while allowing selected Big Data concepts to be explored in later stages.

## Expected Contribution and Outcomes

The expected contribution is a practical demonstration of how multiple data engineering components can be organized into a coherent supply chain monitoring workflow.

The completed project is intended to demonstrate:

- A repeatable workflow for ingesting and preparing supply chain data.

- Structured storage and controlled handling of new, updated, and duplicate records.
- Data quality validation and traceable error handling.
- A planned comparison between relational processing and distributed processing results.
- A dashboard for examining shipment status, delivery delays, and inventory information, subject to the fields available in the selected dataset.
- Clear documentation of the implemented features, limitations, and any optional components that could not be completed.

These outcomes will be assessed against the project's requirements and acceptance criteria after implementation. They are intended objectives, not claims that the platform has already been built or validated.

## Conclusion

The reviewed research establishes the importance of supply chain visibility, data-driven decision-making, and analytical capabilities in modern supply chain management. It also indicates that monitoring and analytics depend on effective integration between operational data and the systems used to process and interpret it.

Building on this background, the proposed project adopts a staged approach that begins with reliable batch processing and relational storage before extending toward distributed processing and dashboard-based monitoring. Its value lies in demonstrating these components as an integrated, testable data engineering workflow within the available project scope.

The literature review therefore provides a rationale for the proposed direction while leaving dataset-specific decisions, implementation details, and performance conclusions to be established during the subsequent project stages.

## References

## References

[1] M. Koot, M. R. K. Mes, and M.-E. Iacob, "A systematic literature review of supply chain decision making supported by the Internet of Things and Big Data Analytics," *Computers & Industrial Engineering*, vol. 154, article 107076, 2021. https://doi.org/10.1016/j.cie.2020.107076.

[2] S. Taj, A. S. Imran, Z. Kastrati, S. M. Daudpota, R. A. Memon, and J. Ahmed, "IoT-based supply chain management: A systematic literature review," *Internet of Things*, vol. 24, article 100982, 2023. https://doi.org/10.1016/j.iot.2023.100982.

[3] I. Lee and G. Mangalaraj, "Big Data Analytics in Supply Chain Management: A Systematic Literature Review and Research Directions," *Big Data and Cognitive Computing*, vol. 6, no. 1, article 17, 2022. https://doi.org/10.3390/bdcc6010017.